

Laws of Boolean Algebra & Logic Gates

Distributive Laws

$$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$$

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

Identity Laws

$$A + 0 = A$$

$$A \cdot 1 = A$$

Null Laws

$$A + 1 = 1$$

$$A \cdot 0 = 0$$

Domination Laws

$$A + 1 = 1$$

$$A \cdot 0 = 0$$

Idempotent Laws

$$A + A = A$$

$$A \cdot A = A$$

Complement Laws

$$A + A' = 1$$

$$A \cdot A' = 0$$

Double Negation Law

$$(A')' = A$$

De Morgan's Laws:

$$\neg(A \cdot B) = \neg A + \neg B$$

$$\neg(A + B) = \neg A \cdot \neg B$$

Commutative Laws

$$A + B = B + A$$

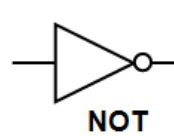
$$A \cdot B = B \cdot A$$

Associative Laws

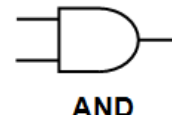
These laws state that:

$$A + (B + C) = (A + B) + C$$

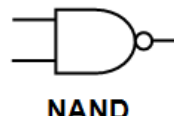
$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$



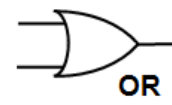
NOT



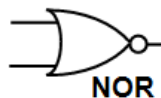
AND



NAND



OR



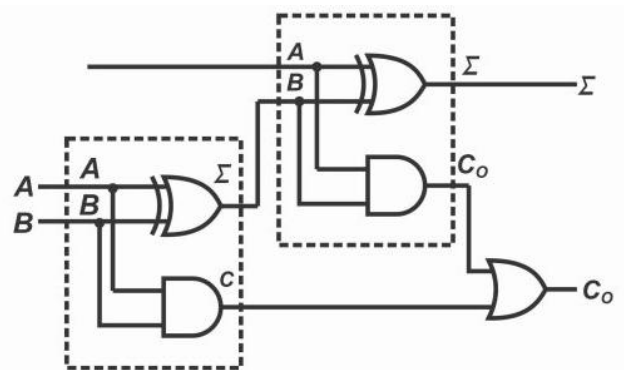
NOR



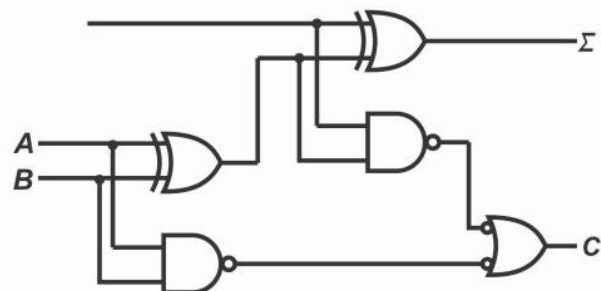
EXCLUSIVE OR



EXCLUSIVE NOR



FULL ADDER



(b)